

## Exercise 2

### Aggregates

①

Q

SELECT DISTINCT department,  
FROM students;

| Department |
|------------|
| IT         |
| HR         |
| Finance    |

Q2

SELECT department,  
AVG(age) AS avg-age  
FROM students  
GROUP BY department;

| department | Avg-age |
|------------|---------|
| IT         | 20.5    |
| HR         | 22.0    |
| Finance    | 23.0    |

Q3

SELECT department,  
(COUNT(\*) AS student\_count  
FROM students  
GROUP BY department  
HAVING COUNT(\*) > 1;

| department | Student Count |
|------------|---------------|
| IT         | 2             |
| HR         | 2             |

Q4

SELECT student\_ID,  
name,  
age,  
department  
FROM students  
WHERE age BETWEEN 21 AND 23;

| Student ID | name    | age | department |
|------------|---------|-----|------------|
| 2          | Bob     | 22  | HR         |
| 3          | Charlie | 21  | IT         |
| 4          | Diana   | 23  | Finance    |
| 5          | Eve     | 22  | HR         |

department  
FROM students  
WHERE age BETWEEN 21 AND 23;

⑤ SELECT student-id,  
name,  
age,  
department

| student-id | name | age | department |
|------------|------|-----|------------|
| 2          | Bob  | 22  | HR         |
| 5          | Eve  | 22  | HR         |

FROM students  
WHERE (department IN ('IT', 'HR')) AND age > 21;

⑥ SELECT department,  
SUM(credits) AS total-credits  
FROM courses  
GROUP BY department  
HAVING SUM(credits) > 5;

| department | Total-credits |
|------------|---------------|
| IT         | 11            |

⑦ SELECT course-id,  
course name,  
department,  
Credits

| Course-id | Course-name | department | Credits |
|-----------|-------------|------------|---------|
| 101       | SQL Basics  | IT         | 3       |
| 104       | Excel       | Finance    | 2       |
| 105       | Statistics  | HR         | 3       |

FROM courses  
WHERE credits < 4;

② ⑤  
 SELECT course-id,  
       course-name,  
       credits

FROM courses  
 ORDER BY credit DESC  
 LIMIT 3;

| Course-id | Course-name  | credits |
|-----------|--------------|---------|
| 102       | Python       | 4       |
| 103       | Data Science | 4       |
| 101       | SQL Basic    | 3       |

④ SELECT MAX(grade) AS max-grade,  
       MIN(grade) AS min-grade,  
       AVG(grade) AS avg-grade  
 FROM enrollments;

| Max-grade | Min-grade | Ave-grade |
|-----------|-----------|-----------|
| 90        | 78        | 84.6      |

⑩ SELECT course-id,  
       COUNT(\*) AS enrollment-count  
 FROM enrollments  
 GROUP BY course-id;

| Course-id | Enrollment Count |
|-----------|------------------|
| 101       | 1                |
| 102       | 1                |
| 103       | 1                |
| 104       | 1                |
| 105       | 1                |

⑪ SELECT department,  
       SUM(salary) AS total-salary,  
       SUM(bonus) AS total-bonus  
 FROM salaries  
 GROUP BY department;

| department | Total salary | Total bonus |
|------------|--------------|-------------|
| IT         | 122000       | 10500       |
| HR         | 109000       | 7500        |
| Finance    | 70000        | 6000        |

② SELECT department,  
 AVG(salary) AS avg-salary  
 FROM salaries  
 GROUP BY department  
 HAVING AVG(salary) > 50000;

| department | Avg-salary |
|------------|------------|
| IT         | 61000      |
| Finance    | 70000      |

③ SELECT employee-id,  
 name,  
 salary,  
 bonus  
 (salary + bonus) AS total-compensation  
 FROM salaries  
 WHERE (salary + bonus) > 60000

| employee-id | name  | salary | bonus | Total-compensation |
|-------------|-------|--------|-------|--------------------|
| 1           | Tom   | 60000  | 5000  | 65000              |
| 3           | Spike | 70000  | 6000  | 76000              |
| 4           | Tyke  | 62000  | 5500  | 67500              |

④ SELECT department,  
 SUM(budget) AS total-budget,  
 AVG(budget) AS avg-budget  
 FROM projects  
 GROUP BY department  
 HAVING (total-budget > 200000);

| department | Total-budget | Avg-budget |
|------------|--------------|------------|
| IT         | 270000       | 135000     |
| Finance    | 80000        | 80000      |

SQL  
SELECT project\_id,  
project\_name,  
department,  
budget

FROM projects

WHERE budget BETWEEN 50000 AND 120000  
AND department <> 'Marketing';